# NUCLEI ON THE ALGEBRA OF TRUTH VALUES AND THE SAMENESS LATTICE OF FUZZY SETS: MEETS, JOINS AND THE FINITE-ITERATION AXIOM

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Abstract. Let $L$ be a complete residuated lattice of truth values and $\mathbf{Set}(L)$ the category of $L$-fuzzy sets. A nucleus $j$ on $L$, a closure operator with $ja \otimes jb \leq j(a \otimes b)$, determines the reflective subcategory of $j$-closed fuzzy sets, with reflector $(X, \mu) \mapsto (X, j\mu)$, and hence the class $W_j$ of maps inverted by the reflector, the isomorphisms up to $j$; these classes are elements of the lattice $\mathrm{Same}(\mathbf{Set}(L))$ of classes of maps closed under composition and two-out-of-three. We show that $j \mapsto W_j$ is an injective map from the lattice of nuclei preserving all meets, and that it preserves the join of two nuclei $j_1, j_2$ exactly when the alternating iterate $(j_2 j_1)^N$ equals $j_1 \vee j_2$ for some $N$, the finite-iteration axiom of the lax logic of the join; in the lattice $\mathrm{Same}(L)$ of the poset $L$ itself, pointwise stabilisation suffices. On a complete MV-algebra every nucleus is closed, $ja = a \vee c$, and on the product chain every nucleus is a closed nucleus or its restriction to the positive degrees; in both cases the alternating iteration stabilises in two steps, and the lattice of nuclei embeds as a lattice into the sameness lattice of fuzzy sets, for MV-algebras as the algebra of truth values itself; the lax modality of Łukasiewicz logic is a disjunction with a constant. On the Gödel chain $[0, 1]$ nuclei are arbitrary inf-closed subsets, and we give pairs for which the join is preserved in $\mathrm{Same}(L)$ but not in $\mathrm{Same}(\mathbf{Set}(L))$, and pairs for which it is preserved in neither.

## 1. Introduction

Fuzzy set theory takes its truth values in a complete residuated lattice $L$ [6, 7]: a fuzzy set is a map $\mu\colon X \to L$, and a map of fuzzy sets $(X, \mu) \to (Y, \nu)$ is a function $f$ with $\mu \leq \nu \circ f$. The resulting category $\mathbf{Set}(L)$ of Goguen [6] is the setting of this paper. The second ingredient is a *nucleus* on $L$, a closure operator $j$ with $ja \otimes jb \leq j(a \otimes b)$ [10, 5]. When $L$ is a frame, nuclei are the Lawvere–Tierney topologies of the topos of sheaves on $L$ and the sublocales of $L$ [8]; in general they are the quotients of $L$ as a residuated lattice and the semantics of the lax modality of intuitionistic and substructural modal logic [4, 15]. A nucleus acts on fuzzy sets degreewise: $(X, \mu) \mapsto (X, j\mu)$ is the reflection of $\mathbf{Set}(L)$ onto the full subcategory of $j$-closed fuzzy sets,

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those with $\mu = j\mu$, and the reflection sends a map $f$ to an isomorphism exactly when $f$ is a bijection with $j\mu = j\nu \circ f$: the fuzzy sets agree once their degrees are rounded up by $j$. We write $W_j$ for this class of maps, the *isomorphisms up to* $j$.

The classes $W_j$ live in the lattice $\mathrm{Same}(\mathbf{Set}(L))$ of *samenesses* of $\mathbf{Set}(L)$, the classes of maps containing the identities and closed under composition and two-out-of-three, studied in [11, 12, 13, 14] for Grothendieck topologies and for subvarieties. The question of those papers is how the lattice operations of the indexing structure, here the lattice $\mathrm{N}(L)$ of nuclei, are reflected in Same. The answer here is complete.

**Theorem 1.1.** *Let $L$ be a complete residuated lattice.*

(i) *$j \mapsto W_j$ is an injective, order-preserving map $\mathrm{N}(L) \to \mathrm{Same}(\mathbf{Set}(L))$ that preserves all meets: $W_{\bigwedge_i j_i} = \bigcap_i W_{j_i}$ for every family of nuclei.*

(ii) *For nuclei $j_1, j_2$ preserving binary meets, $W_{j_1 \vee j_2} = W_{j_1} \vee W_{j_2}$ if and only if there is $N$ with $(j_2 j_1)^N = j_1 \vee j_2$.*

*The same holds for the classes $W_j^L = \{a \leq b : \ ja = jb\}$ in the sameness lattice $\mathrm{Same}(L)$ of the poset $L$, except that in (ii) the condition is pointwise: for every $a \in L$ some iterate $(j_2 j_1)^n a$ equals $(j_1 \vee j_2)a$.*

Every nucleus on a Heyting algebra, every nucleus on an MV-algebra and every closed nucleus $a \mapsto a \vee c$ on a distributive lattice preserves binary meets, so (ii) covers the standard algebras of fuzzy logic. The condition in (ii) is the *finite-iteration axiom* $(\bigcirc_2 \bigcirc_1)^N \varphi \leftrightarrow \bigcirc \varphi$ of the trimodal lax logic **LJ** of two nuclei and their join [15], valid on every finite algebra and on $\mathrm{Up}(P)$ for finite posets $P$ with $N = |P|$; Theorem 1.1 says that this axiom is exactly what makes the join of two nuclei visible at the level of isomorphisms of fuzzy sets. The gap between the pointwise and the uniform condition separates $\mathrm{Same}(L)$ from $\mathrm{Same}(\mathbf{Set}(L))$.

For the standard algebras the picture is as follows (Section 5). On a complete MV-algebra $A$, in particular on the Łukasiewicz chain $[0,1]$, every nucleus is closed, $ja = a \vee c$: the fixed points of a nucleus are closed under $a \to b$, and $\{a \to b : a \in A\}$ is the whole principal filter $\uparrow b$. Hence $\mathrm{N}(A) \cong A$, two alternating steps always suffice, and $c \mapsto W_{a \mapsto a \vee c}$ is a lattice embedding $A \hookrightarrow \mathrm{Same}(\mathbf{Set}(A))$: the algebra of truth values is a sublattice of the sameness lattice of its fuzzy sets, $W_c$ being the class of bijections under which the two fuzzy sets agree above the threshold $c$. On the product chain $[0,1]_\Pi$ the residual $a \to 0$ takes only the values 0 and 1, and besides the closed nuclei there are their restrictions to the positive degrees, $q_c(a) = a \vee c$ for $a > 0$ and $q_c(0) = 0$; still two alternating steps suffice and $\mathrm{N}([0,1]_\Pi)$ embeds as a lattice. On the Gödel chain $[0,1]$, by contrast, a nucleus is any inf-closed set of fixed points containing 1, and we give two nuclei whose alternating iteration stabilises at every point but not uniformly, so that the join is preserved in $\mathrm{Same}(L)$ and not in $\mathrm{Same}(\mathbf{Set}(L))$, and two nuclei for which it is preserved in neither.

**What is known.** Nuclei on frames, quantales and residuated lattices and their lattice are classical [8, 10, 5]; the join of two nuclei is the least common fixed point operator, reached by transfinite iteration of the composite [9, 3]. Goguen's category of $L$-fuzzy sets is from [6]; its relation to sheaves on a frame is due to Barr [1]. Samenesses and the behaviour of meets and joins of Grothendieck topologies and of subvarieties in Same are from [11, 12, 13, 14], and the lax logic **LJ** of the join of two nuclei, with its finite-iteration axiom, from [15]. Nuclei on residuated lattices are studied in [5], and a description of arbitrary nuclei on residuated lattices, with the terms that define them on subvarieties of BL-algebras, is given by Buss, Castaño and Díaz Varela [2]. That nuclei on MV-algebras are closed is elementary and may be known; we include the short proof, and we classify the nuclei on the product chain.

**What is new.** Theorem 1.1: the meet-embedding $\mathrm{N}(L) \hookrightarrow \mathrm{Same}(\mathbf{Set}(L))$ and the characterisation of the preservation of binary joins by the finite-iteration axiom, with the pointwise variant for $\mathrm{Same}(L)$; the description of $\mathrm{Same}(\mathbf{Set}(L))$-joins by uniformly bounded alternating chains (Lemma 4.1); and the consequences for the standard algebras, the lattice embeddings of $\mathrm{N}(A)$ for complete MV-algebras and for the product chain and the two Gödel examples (Section 5); the classification of nuclei on MV-algebras and on the product chain is elementary.

## 2. Fuzzy sets, nuclei and isomorphisms up to a nucleus

Throughout, $L$ is a complete integral commutative residuated lattice: a complete lattice with a commutative monoid operation $\otimes$ with unit $\top$, distributing over arbitrary joins, and residual $\to$ with $a \otimes b \leq c$ iff $a \leq b \to c$. Frames ($\otimes = \wedge$, Gödel logic), MV-algebras (Łukasiewicz logic) and the product chain are examples [7].

**Definition 2.1.** A *nucleus* on $L$ is a map $j\colon L \to L$ with $a \leq ja$, $jja = ja$, $a \leq b \Rightarrow ja \leq jb$ and $ja \otimes jb \leq j(a \otimes b)$. Its fixed points $\mathrm{Fix}(j) = \{a : ja = a\}$ are closed under arbitrary meets and under $a \to b$ for $b \in \mathrm{Fix}(j)$; conversely every subset $F \subseteq L$ closed under meets and under $a \to (-)$ is $\mathrm{Fix}(j_F)$ for the nucleus $j_F(a) = \bigwedge\{f \in F : f \geq a\}$ [10, 5]. Nuclei are ordered pointwise; $\mathrm{N}(L)$ is a complete lattice with pointwise meets, $j \leq k$ iff $\mathrm{Fix}(k) \subseteq \mathrm{Fix}(j)$, and the join $j_1 \vee j_2$ is the nucleus with $\mathrm{Fix}(j_1 \vee j_2) = \mathrm{Fix}(j_1) \cap \mathrm{Fix}(j_2)$, i.e. $(j_1 \vee j_2)(a)$ is the least common fixed point above $a$, the stable value of the transfinite iteration $a \leq j_1 a \leq j_2 j_1 a \leq \cdots$ [9, 3]. A nucleus *preserves binary meets* if $j(a \wedge b) = ja \wedge jb$; this holds for every nucleus on a frame, where it is part of the definition, and for closed nuclei $c_u(a) = a \vee u$ on distributive lattices. A closure operator always satisfies $j(a \vee b) = j(ja \vee jb)$.

**Definition 2.2.** $\mathbf{Set}(L)$ has as objects the pairs $(X, \mu)$ with $X$ a set and $\mu\colon X \to L$, and as morphisms $(X, \mu) \to (Y, \nu)$ the functions $f\colon X \to Y$ with $\mu(x) \leq \nu(f(x))$ for all $x$. An isomorphism is a bijection with $\mu = \nu \circ f$.

For a nucleus $j$, a fuzzy set is *$j$-closed* if $\mu = j\mu$, and $\mathbf{Set}_j(L)$ is the full subcategory of $j$-closed fuzzy sets.

**Lemma 2.3.** *$\mathbf{Set}_j(L)$ is reflective in $\mathbf{Set}(L)$ with reflector $(X,\mu) \mapsto (X, j\mu)$ and unit the identity function. The reflector inverts a map $f\colon (X,\mu) \to (Y,\nu)$ if and only if $f$ is a bijection and $j\mu = j\nu \circ f$. $\mathbf{Set}_{j_1}(L) \cap \mathbf{Set}_{j_2}(L) = \mathbf{Set}_{j_1 \vee j_2}(L)$, and $j \le k$ iff $\mathbf{Set}_k(L) \subseteq \mathbf{Set}_j(L)$.*

*Proof.* If $(Z,\zeta)$ is $j$-closed and $f\colon (X,\mu) \to (Z,\zeta)$, then $j\mu \le j(\zeta \circ f) = \zeta \circ f$, so $f$ is a map $(X, j\mu) \to (Z,\zeta)$, uniquely through the unit. The reflector sends $f$ to the same function regarded as $(X, j\mu) \to (Y, j\nu)$, which is an isomorphism iff $f$ is bijective and $j\mu = j\nu \circ f$. A fuzzy set is closed under $j_1$ and $j_2$ iff its degrees are common fixed points, i.e. fixed by $j_1 \vee j_2$. If $j \le k$ then $k\mu = \mu$ implies $j\mu \le k\mu = \mu$; if $\mathbf{Set}_k \subseteq \mathbf{Set}_j$ then every $k$-fixed element is $j$-fixed, so $j \le k$. □

**Definition 2.4.** For a category $C$, $\mathrm{Same}(C)$ is the complete lattice of classes of morphisms containing all identities and closed under composition and two-out-of-three (if two of $f$, $g$, $g \circ f$ are in the class, so is the third), ordered by inclusion; meets are intersections and the join of two classes is the least class containing both [11]. For a nucleus $j$ let

$$W_j = \{f\colon (X,\mu) \to (Y,\nu) \text{ bijective with } j\mu = j\nu \circ f\} \in \mathrm{Same}(\mathbf{Set}(L)),$$

the class inverted by the reflector of Lemma 2.3, which is a sameness because classes inverted by a functor are. In the poset $L$, viewed as a category with an arrow $a \to b$ for $a \le b$, let $W_j^L = \{(a \le b) :\ ja = jb\} \in \mathrm{Same}(L)$, the class inverted by the monotone map $j\colon L \to L$.

The class Bij of maps of fuzzy sets that are bijective on underlying sets is a sameness, and $W_j \subseteq \mathrm{Bij}$. A bijective map is, up to composition with isomorphisms, an identity function $(X,\mu) \to (X,\nu)$ with $\mu \le \nu$ pointwise; since isomorphisms belong to every sameness and samenesses are closed under composition and cancellation with them, membership of a bijection in a sameness depends only on the family of pairs $(\mu(x) \le \nu(f(x)))_{x \in X}$. In this sense the whole discussion is pointwise in $L$, and the only global feature is uniformity over $X$.

## 3. Meets

**Theorem 3.1.** *For any family $(j_i)_{i \in I}$ of closure operators on $L$ with pointwise meet $j = \bigwedge_i j_i$,*

$$W_j = \bigcap_i W_{j_i} \quad \text{in } \mathrm{Same}(\mathbf{Set}(L)), \qquad W_j^L = \bigcap_i W_{j_i}^L \quad \text{in } \mathrm{Same}(L).$$

*Moreover $j \mapsto W_j$ and $j \mapsto W_j^L$ are injective and order-preserving. Hence both are embeddings of complete meet-semilattices.*

*Proof.* If $j \leq k$ then $kj = k$, so $j\mu = j\nu f$ gives $k\mu = kj\mu = kj\nu f = k\nu f$: $W_j \subseteq W_k$. Injectivity: the pair $(a \leq ja)$ lies in $W_j^L$, and in $W_k^L$ iff $ka = kja$, i.e. $ja \leq ka$; so $W_j^L = W_k^L$ forces $j = k$, and the same pairs realised on one-point fuzzy sets give injectivity of $j \mapsto W_j$. For the meets, $\supseteq$ is monotonicity. Conversely let $f\colon (X,\mu) \to (Y,\nu)$ be a bijection with $j\mu = j\nu f$ and fix $x$, $a = \mu(x) \leq b = \nu(f(x))$. Then

$$b \leq jb = \textstyle\bigwedge_i j_i b = \bigwedge_i j_i a \leq j_k a$$

for every $k$, so $j_k b \leq j_k j_k a = j_k a \leq j_k b$. Hence $j_k\mu = j_k\nu f$ for all $k$. The argument for $\mathrm{Same}(L)$ is the case of one point. □

Theorem 3.1 uses nothing about $\otimes$: it holds for all closure operators on a complete lattice. The pointwise meet of nuclei is their meet in $\mathrm{N}(L)$, so the theorem is statement (i) of Theorem 1.1.

## 4. Joins

Fix two nuclei $j_1, j_2$ on $L$ preserving binary meets, and write $j = j_1 \vee j_2$. An *alternating chain* from $a$ to $b$ is a sequence $a = c_0 \leq c_1 \leq \cdots \leq c_n = b$ together with indices $p(1), \ldots, p(n) \in \{1,2\}$ such that $j_{p(i)}c_{i-1} = j_{p(i)}c_i$ for all $i$; its length is $n$. Repeating an element is a step of either index, so chains can be padded, and the indices can be assumed to alternate $1,2,1,2,\ldots$ by inserting trivial steps; we do so.

**Lemma 4.1.** (i) *$W_{j_1}^L \vee W_{j_2}^L$ in $\mathrm{Same}(L)$ is the set of pairs $a \leq b$ joined by an alternating chain.*

(ii) *$W_{j_1} \vee W_{j_2}$ in $\mathrm{Same}(\mathbf{Set}(L))$ is the set of bijections $f\colon (X,\mu) \to (Y,\nu)$ for which there is $N$ such that every pair $\mu(x) \leq \nu(f(x))$ is joined by an alternating chain of length at most $N$.*

*Proof.* Let $\mathcal{U}$ be the class in (ii); $\mathcal{U} \supseteq W_{j_1} \cup W_{j_2}$ (chains of length 1) and $\mathcal{U}$ contains the identities. Each element of $\mathcal{U}$ is a composite of elements of $W_{j_1} \cup W_{j_2}$: given uniform chains $(c_i(x))_{i \leq N}$ with alternating indices, the identity functions $(X, c_{i-1}) \to (X, c_i)$ lie in $W_{j_{p(i)}}$, and $f$ is the composite of these with the isomorphism $(X, \nu f) \to (Y, \nu)$. So it remains to show that $\mathcal{U}$ is a sameness. Composition concatenates chains, with lengths adding. For cancellation let $f\colon (X,\mu) \to (Y,\nu)$ and $g\colon (Y,\nu) \to (Z,\rho)$ with $gf \in \mathcal{U}$; both are bijections if one of $f, g$ is in $\mathcal{U}$, as Bij is a sameness. Fix $x$ and put $a = \mu(x) \leq b = \nu(fx) \leq c = \rho(gfx)$, with a chain $a = c_0 \leq \cdots \leq c_n = c$ of length at most $N$. If $g \in \mathcal{U}$ we need a bounded chain from $a$ to $b$: $a = c_0 \wedge b \leq c_1 \wedge b \leq \cdots \leq c_n \wedge b = b$ is one, because $j_{p(i)}(c_{i-1} \wedge b) = j_{p(i)}c_{i-1} \wedge j_{p(i)}b = j_{p(i)}c_i \wedge j_{p(i)}b = j_{p(i)}(c_i \wedge b)$ by preservation of binary meets. If $f \in \mathcal{U}$ we need a bounded chain from $b$ to $c$: $b = c_0 \vee b \leq c_1 \vee b \leq \cdots \leq c_n \vee b = c$ is one, because $j(c_{i-1} \vee b) = j(jc_{i-1} \vee jb) = j(jc_i \vee jb) = j(c_i \vee b)$ for $j = j_{p(i)}$, using $j(u \vee v) = j(ju \vee jv)$. Both constructions preserve the length, hence the uniform bound. This proves (ii), and (i) is the same argument on one point, where no bound is needed. □

**Remark 4.2.** Preservation of binary meets is used only in the left cancellation, and cannot simply be dropped from Lemma 4.1 for closure operators in general. On the four-element Boolean lattice $\{0, a, b, 1\}$ let $j_1$ fix $0, a, 1$ and send $b$ to 1, and let $j_2$ fix $b, 1$ and send 0 to $b$ and $a$ to 1; $j_1$ does not preserve $a \wedge b$. The pairs $(0 \le b)$, $(b \le 1)$, $(a \le 1)$ are steps, so $(0 \le 1)$ and $(a \le 1)$ are joined by chains, but $(0 \le a)$ is not, since the only step out of 0 leads to $b$; the chained pairs do not form a sameness. (These are closure operators, not nuclei for $\otimes = \wedge$.) Whether Theorem 4.3 itself extends to nuclei that do not preserve binary meets is left open.

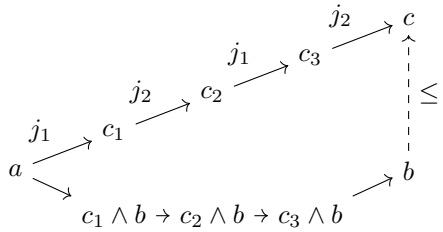



FIGURE 1. Left cancellation in Lemma 4.1: an alternating chain from $a$ to $c$ is cut down to one from $a$ to $b \le c$ by meeting with $b$; each step keeps its index because the nuclei preserve binary meets.

**Theorem 4.3.** *Let $j_1, j_2$ be nuclei preserving binary meets, $j = j_1 \vee j_2$ and $S = j_2 j_1$.*

(i) $W^L_{j_1} \vee W^L_{j_2} = W^L_j$ *in* $\mathrm{Same}(L)$ *if and only if for every $a \in L$ there is $n$ with $S^n a = ja$.*

(ii) $W_{j_1} \vee W_{j_2} = W_j$ *in* $\mathrm{Same}(\mathbf{Set}(L))$ *if and only if there is $N$ with $S^N = j$.*

*In both cases the inclusion $\subseteq$ always holds.*

*Proof.* The inclusion holds since $W_{j_1}, W_{j_2} \subseteq W_j$ by monotonicity. An alternating chain $a = c_0 \le \cdots \le c_n = b$ satisfies $c_i \le j_{p(i)} c_{i-1}$; padding it with a trivial first step if it starts with index 2, so that the indices read $1, 2, 1, 2, \ldots$, gives $b \le S^{\lceil (n+1)/2 \rceil} a \le S^{n+1} a$. Conversely $a \le j_1 a \le j_2 j_1 a \le \cdots \le S^n a$ is an alternating chain of length $2n$.

(i) If every $a$ stabilises, let $a \le b$ with $ja = jb$. There are chains from $a$ to $ja$ and from $b$ to $jb = ja$, so $(a \le ja)$ and $(b \le jb)$ lie in the join by Lemma 4.1, and $(a \le b)$ does by cancellation. Conversely, $(a \le ja) \in W^L_j$; if it lies in the join there is an alternating chain from $a$ to $ja$ of some length $n$, so $ja \le S^{n+1} a \le ja$.

(ii) If $S^N = j$, every pair $a \le ja$ has a chain of length $2N$, and for a bijection $f$ with $j\mu = j\nu f$ the pairs $(\mu(x) \le j\mu(x))$ and $(\nu(fx) \le j\nu(fx))$ are uniformly chained, so $f$ is in the join by Lemma 4.1 and cancellation,

which preserves bounds. Conversely suppose no $N$ works: for every $k$ there is $a_k$ with $S^{k+1}a_k \neq ja_k$, so every alternating chain from $a_k$ to $ja_k$ has length $> k$ by the first paragraph. Let $X = \mathbb{N}$, $\mu(k) = a_k$, $\nu(k) = ja_k$. The identity $(X, \mu) \to (X, \nu)$ lies in $W_j$ but is not uniformly chained, so it is not in $W_{j_1} \vee W_{j_2}$. □

**Corollary 4.4.** *For nuclei preserving binary meets, $W_{j_1 \vee j_2} = W_{j_1} \vee W_{j_2}$ in* $\mathrm{Same}(\mathbf{Set}(L))$ *if and only if the algebra $(L, j_1, j_2, j_1 \vee j_2)$ validates the finite-iteration axiom $(\bigcirc_2 \bigcirc_1)^N \varphi \leftrightarrow \bigcirc \varphi$ of* **LJ** *for some $N$* [15]*. In particular this holds for every finite $L$, for $L = \mathrm{Up}(P)$ with $P$ a finite poset ($N = |P|$), and whenever $\mathrm{N}(L)$ satisfies the ascending chain condition on $\{S^n : n \in \mathbb{N}\}$.*

**Remark 4.5.** The join of samenesses is computed by chains of *bounded* length and the join of nuclei by transfinite iteration; Theorem 4.3 is the statement that the two agree exactly when the iteration is uniformly finite. This is the mechanism of [12, 13] for Grothendieck topologies, where the alternating products of two idempotent ideals must stabilise, reduced to its degreewise core: the category of fuzzy sets has no gluing, so nothing but the arithmetic of $L$ is involved, and the pointwise condition of (i) is the residue of (ii) on a single point.

## 5. The standard algebras

**Proposition 5.1.** *Let $A$ be a complete MV-algebra. Then every nucleus on $A$ is closed, $j(a) = a \vee c$ with $c = j(\bot)$, and $c \mapsto (-) \vee c$ is a lattice isomorphism $A \cong \mathrm{N}(A)$. Consequently $j_2 j_1 = j_1 \vee j_2$ for all nuclei, and $c \mapsto W_{(-)\vee c}$ is a lattice embedding $A \hookrightarrow \mathrm{Same}(\mathbf{Set}(A))$ and $A \hookrightarrow \mathrm{Same}(A)$, where $W_{(-)\vee c}$ is the class of bijections $f$ with $\mu \vee c = (\nu \circ f) \vee c$.*

*Proof.* Let $j$ be a nucleus and $b \in \mathrm{Fix}(j)$. Then $a \to b \in \mathrm{Fix}(j)$ for every $a$. In an MV-algebra $a \to b = \neg a \oplus b$, and for every $x \geq b$ we have $b \oplus (x \otimes \neg b) = b \vee x = x$, so $x = a \to b$ for $a = \neg(x \otimes \neg b)$; hence $\uparrow b \subseteq \mathrm{Fix}(j)$. So $\mathrm{Fix}(j)$ is an up-set closed under meets, i.e. $\mathrm{Fix}(j) = \uparrow c$ with $c = \bigwedge \mathrm{Fix}(j) = j(\bot)$, and $j(a) = a \vee c$. Conversely $a \mapsto a \vee c$ is a nucleus: $(a \vee c) \otimes (b \vee c) = (a \otimes b) \vee (a \otimes c) \vee (c \otimes b) \vee (c \otimes c) \leq (a \otimes b) \vee c$ by integrality. The map $c \mapsto (-) \vee c$ is an order isomorphism $A \to \mathrm{N}(A)$, with $\mathrm{Fix}((-) \vee c) \cap \mathrm{Fix}((-) \vee d) = \uparrow (c \vee d)$ and pointwise meet $(-) \vee (c \wedge d)$ by distributivity. For the iteration, $j_2 j_1 a = a \vee c \vee d$ is already fixed by both. Closed nuclei on a distributive lattice preserve binary meets, so Theorems 3.1 and 4.3 apply with $N = 1$, and injectivity is Theorem 3.1. □

**Proposition 5.2.** *On the product chain $[0,1]_\Pi$ ($a \otimes b = ab$, $a \to b = \min(1, b/a)$ for $a > 0$, $0 \to b = 1$) the nuclei are exactly the closed nuclei $p_c(a) = a \vee c$, $c \in [0,1]$, and the maps $q_c$, $c \in (0,1]$, with $q_c(0) = 0$ and $q_c(a) = a \vee c$ for $a > 0$. All of them preserve binary meets, $(j_2 j_1)^2 = j_1 \vee j_2$ for any two of them, and $j \mapsto W_j$ is a lattice embedding $\mathrm{N}([0,1]_\Pi) \hookrightarrow \mathrm{Same}(\mathbf{Set}([0,1]_\Pi))$.*

*Proof.* Let $j$ be a nucleus. If $b > 0$ is fixed then so is $a \to b = b/a$ for $a \in [b, 1]$, which runs through $[b, 1]$; so $\mathrm{Fix}(j) \cap (0, 1]$ is an up-set of $(0, 1]$ closed under infima, hence $(0, 1]$, $[c, 1]$ with $c > 0$, or (as 1 is fixed) never empty. The value 0 is fixed or not, and $a \to 0 \in \{0, 1\}$ imposes nothing. If $\mathrm{Fix}(j) \cap (0, 1] = (0, 1]$ then 0 is fixed as the infimum, giving $p_0 = \mathrm{id}$; otherwise $\mathrm{Fix}(j) = [c, 1]$ or $\{0\} \cup [c, 1]$ with $c > 0$, giving $p_c$ or $q_c$. Conversely $p_c$ is a nucleus as in Proposition 5.1, and $q_c$ is one: $q_c(a)q_c(b) = 0 \leq q_c(ab)$ if $a = 0$ or $b = 0$, and otherwise $(a \vee c)(b \vee c) = \max(ab, ac, cb, c^2) \leq \max(ab, c) = q_c(ab)$. Both kinds preserve binary meets, the chain being distributive and $q_c$ agreeing with $p_c$ on $(0, 1]$ and fixing 0. For the iteration, on $a > 0$ every composite of two of these nuclei is $a \vee c \vee d$, fixed by both; at 0 the only nontrivial case is $j_1 = q_c$, $j_2 = p_d$ with $d > 0$, where $j_2 j_1(0) = d$ and $(j_2 j_1)^2(0) = d \vee c = (j_1 \vee j_2)(0)$, as $\mathrm{Fix}(q_c) \cap \mathrm{Fix}(p_d) = [c \vee d, 1]$. So $N = 2$ always works and Theorem 4.3 gives the embedding, with Theorem 3.1 for meets and injectivity. Note that $\mathrm{N}([0, 1]_\Pi)$ is not isomorphic to $[0, 1]$: it is the chain of the $p_c$ with a second copy $q_c < p_c$ of its positive part attached. □

**Remark 5.3.** In the language of [15], the lax modalities of Łukasiewicz logic with truth constants are the disjunctions $\bigcirc\varphi = \varphi \vee c$, and the join of two of them is the disjunction with $c \vee d$: the trimodal logic **LJ** over Łukasiewicz logic collapses to its finite-iteration instance with $N = 1$. Over Gödel logic, where nuclei are the Heyting ones, no such collapse occurs, as the next examples show.

**Example 5.4.** Let $L = [0, 1]$ with the Gödel operations, $\otimes = \min$ and $a \to b = 1$ if $a \leq b$, $= b$ otherwise. Since $\{a \to b : a \in L\} = \{b, 1\}$, the fixed-point sets of nuclei are exactly the subsets $F \ni 1$ closed under infima, with $j_F(a) = \min\{f \in F : f \geq a\}$; all nuclei preserve binary meets.

(a) *Pointwise but not uniformly.* Let $F_1 = \{0, 1\} \cup \{1/(2k) : k \geq 1\}$ and $F_2 = \{0, 1\} \cup \{1/(2k+1) : k \geq 1\}$, both inf-closed. Then $\mathrm{Fix}(j_1 \vee j_2) = F_1 \cap F_2 = \{0, 1\}$, so $j(0) = 0$ and $j(a) = 1$ for $a > 0$. From $a = 1/n$ the alternating iteration moves $1/n \mapsto 1/(n-1) \mapsto \cdots \mapsto 1$, with $n - 1$ nontrivial steps (the first step is trivial when $1/n \in F_1$), and 0 is fixed; so every point stabilises but no $N$ works. By Theorem 4.3, $W^L_{j_1} \vee W^L_{j_2} = W^L_j$ in $\mathrm{Same}([0, 1]_G)$ while $W_{j_1} \vee W_{j_2} \subsetneq W_j$ in $\mathrm{Same}(\mathbf{Set}([0, 1]_G))$: the identity $(\mathbb{N}, n \mapsto 1/n) \to (\mathbb{N}, n \mapsto 1)$ is an isomorphism up to $j$ but not a bounded alternation of isomorphisms up to $j_1$ and up to $j_2$.

(b) *Not even pointwise.* Let $F_1 = \{1\} \cup \{1 - 1/(2k) : k \geq 1\}$ and $F_2 = \{1\} \cup \{1 - 1/(2k+1) : k \geq 1\}$. Then $F_1 \cap F_2 = \{1\}$, so $j$ is the constant 1, while the iteration from 0 runs through $1/2, 2/3, 3/4, \ldots$ and never reaches 1. Hence $(0 \leq 1) \in W^L_j$ is in neither join: $W^L_{j_1} \vee W^L_{j_2} \subsetneq W^L_j$ already in $\mathrm{Same}([0, 1]_G)$. This is the Gödel-chain form of the inexpressibility example of [15, §5].

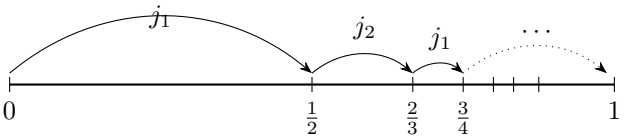



FIGURE 2. Example 5.4(b) on the Gödel chain $[0,1]$: the alternating iteration from 0 climbs $1-1/k$ and never reaches $(j_1 \vee j_2)(0) = 1$.

**Example 5.5.** For $L = \mathrm{Up}(P)$, the up-sets of a poset, the nuclei $\bigcirc_C U = \{x :\uparrow x \cap C \subseteq U\}$ of [15, 16] preserve binary meets. For finite $P$ every nucleus is of this form and $(j_2 j_1)^{|P|} = j_1 \vee j_2$ [15, Thm. 4.1], so $\mathrm{N}(\mathrm{Up}(P)) \cong (\mathcal{P}(P), \supseteq)$ embeds as a lattice into $\mathrm{Same}(\mathbf{Set}(\mathrm{Up}(P)))$. For infinite $P$ the join $\bigcirc_C \vee \bigcirc_{C'}$ equals $\bigcirc_{C \cap C'}$ iff $P$ has no $(C \cap C')$-free alternating chain [16], a different condition from the stabilisation of the iteration; Example 5.4(a) and (b) are modelled on $\mathrm{Up}(\omega + 1)$ and $\mathrm{Up}(\omega + 2)$ with the even and odd points, whose up-sets form chains.

## 6. Remarks on the logical reading

A triple $(L, j_1, j_2, j_1 \vee j_2)$ is an algebra of the lax logic **LJ** of two nuclei and their join [15], and fuzzy sets over $L$ are its natural models: a fuzzy set $(X, \mu)$ assigns to each point a degree, the modalities act on degrees, and $W_j$ consists of the maps along which the $j$-modalised degrees are transported isomorphically. Theorem 4.3 then reads: the isomorphisms up to the join modality are generated, by composition and two-out-of-three, by the isomorphisms up to the two modalities if and only if the join modality is a finite iterate of the two, the finite-iteration axiom. Over Łukasiewicz logic the axiom holds with $N = 1$ and the modalities are disjunctions with constants (Proposition 5.1), over product logic with $N = 2$ (Proposition 5.2); over Gödel logic it may fail (Example 5.4). The other operations of $\mathbf{Set}(L)$, in particular the $\alpha$-cuts $X_\alpha = \{x : \mu(x) \geq \alpha\}$, give samenesses $W_\alpha = \{f : f \text{ restricts to a bijection } X_\alpha \to Y_\alpha\}$ that are not of the form $W_j$ and do not preserve joins of levels ($W_\alpha \cap W_\beta \not\subseteq W_{\alpha\vee\beta}$ on $L = \mathbf{2} \times \mathbf{2}$); the structure of $\mathrm{Same}(\mathbf{Set}(L))$ beyond the image of $\mathrm{N}(L)$ is left open.

**Question 6.1.** For which $L$ is every element of $\mathrm{Same}(\mathbf{Set}(L))$ contained in Bij generated by the classes $W_j$ and the cut classes $W_\alpha$? Does Theorem 4.3(ii) extend to nuclei that do not preserve binary meets?

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## References


[1] M. Barr, *Fuzzy set theory and topos theory*, Canad. Math. Bull. **29** (1986), 501–508.

[2] S. Buss, D. Castaño, J. P. Díaz Varela, *Terms that define nuclei on residuated lattices: a case study of BL-algebras*, preprint, arXiv:2412.19690, 2024.

[3] M. H. Escardó, *Joins in the frame of nuclei*, Appl. Categ. Structures **11** (2003), 117–124.

[4] M. Fairtlough, M. Mendler, *Propositional lax logic*, Inform. and Comput. **137** (1997), 1–33.

[5] N. Galatos, P. Jipsen, T. Kowalski, H. Ono, *Residuated Lattices: An Algebraic Glimpse at Substructural Logics*, Elsevier, 2007.

[6] J. A. Goguen, *L-fuzzy sets*, J. Math. Anal. Appl. **18** (1967), 145–174.

[7] P. Hájek, *Metamathematics of Fuzzy Logic*, Kluwer, 1998.

[8] P. T. Johnstone, *Stone Spaces*, Cambridge University Press, 1982.

[9] D. S. Macnab, *Modal operators on Heyting algebras*, Algebra Universalis **12** (1981), 5–29.

[10] K. I. Rosenthal, *Quantales and Their Applications*, Longman, 1990.

[11] R. Shibusawa, *The sameness lattice of a category: weak-equivalence structures, Galois functoriality, and comparison intervals*, preprint, 2026, DOI 10.5281/zenodo.22823772.

[12] R. Shibusawa, *Meets and joins of Grothendieck topologies in the sameness lattice*, preprint, 2026, DOI 10.5281/zenodo.22978446.

[13] R. Shibusawa, *Grothendieck topologies on a finite category form a sublattice of the lattice of two-out-of-three classes*, preprint, 2026, DOI 10.5281/zenodo.23204943.

[14] R. Shibusawa, *Lattices of subvarieties and of logics as sublattices of the lattice of two-out-of-three classes*, preprint, 2026, DOI 10.5281/zenodo.23407364.

[15] R. Shibusawa, *The join of two nuclei as a lax modality: axiomatization, finite models, and inexpressibility*, preprint, 2026, DOI 10.5281/zenodo.23330576.

[16] R. Shibusawa, *Joins of relativized necessities on a poset: an alternating-chain criterion and the induction scheme of lax logic*, preprint, 2026, DOI 10.5281/zenodo.23411203.



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